

Nigeria Development Indicators Analysis

A Data-Driven Assessment of Population, Economic, Infrastructure, Health, and Environmental Trends

1. Objective / Problem Statement

Objective

The objective of this project is to analyze Nigeria's long-term development trajectory using selected indicators from the World Bank's World Development Indicators (WDI) dataset.

The analysis examines trends in Nigeria's population, economic performance, electricity access, life expectancy, forest coverage, and carbon dioxide emissions. Python was used for data cleaning, preparation, exploratory data analysis, and calculation of derived metrics, while Power BI was used to create an interactive dashboard for visualizing the results.

The project aims to identify major development trends, areas of progress, and areas requiring continued attention.

Problem Statement

Nigeria has experienced significant demographic, economic, infrastructural, health, and environmental changes over several decades. However, examining individual statistics separately does not provide a complete understanding of the country's overall development trajectory.

This project therefore investigates the following question:

How have key development indicators in Nigeria changed over time, and what do these changes reveal about the country's demographic growth, economic development, infrastructure access, population well-being, and environmental sustainability?

The analysis focuses on:

- Population growth and demographic pressure.
 - Changes in GDP per capita and economic performance.
 - Progress in access to electricity.
 - Changes in life expectancy.
 - Changes in forest coverage.
 - Changes in CO₂ emissions per capita.
 - Comparison of the long-term percentage change across the selected indicators.
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2. Dataset Description

Data Source

The dataset used for this project was obtained from the **World Bank World Development Indicators (WDI)** database.

The World Development Indicators is a comprehensive collection of global development statistics covering more than 200 countries and territories. It includes indicators across areas such as:

- Economy
- Population
- Health
- Education
- Environment
- Infrastructure
- Trade
- Technology
- Governance

Dataset Scope

The original dataset contained development information for countries around the world. For this project, the dataset was filtered to focus exclusively on **Nigeria**.

The final cleaned dataset used for the analysis contained:

- **451 rows**
- **6 columns**
- **0 missing values in the** `Value` **column**

The selected indicators covered several dimensions of national development.

Development Area	Indicator
Demographics	Population, total
Demographics	Population growth (annual %)
Economy	GDP per capita (current US\$)
Economy	GDP growth (annual %)
Infrastructure	Access to electricity (% of population)
Health	Life expectancy at birth, total (years)
Environment	Forest area (% of land area)
Environment	CO ₂ emissions (metric tons per capita)

The indicators do not all cover the same time period because data availability varies by indicator. Some indicators have records beginning in 1960, while others begin in later years.

3. Data Cleaning Process

The raw World Bank dataset contained information for multiple countries and numerous development indicators. Data cleaning and preparation were therefore performed before analysis.

3.1 Filtering the Dataset

The dataset was filtered to include only records where the country was **Nigeria**.

This removed data for other countries and created a country-specific dataset for the analysis.

3.2 Selecting Relevant Indicators

Eight indicators were selected based on their relevance to the project's focus on Nigeria's demographic, economic, infrastructural, health, and environmental development.

The selected indicators were:

- Population, total
- Population growth (annual %)
- GDP per capita (current US\$)
- GDP growth (annual %)
- Access to electricity (% of population)
- Life expectancy at birth, total (years)
- Forest area (% of land area)
- CO₂ emissions (metric tons per capita)

3.3 Removing Unnecessary Columns

Unnecessary fields from the original dataset were removed so that the final dataset contained only the columns required for analysis.

The final dataset contained:

- Country
- Indicator Name
- Year
- Value
- Unit
- Indicator Code

3.4 Handling Missing Values

Missing values were assessed after filtering the dataset.

The final analysis dataset contained **no missing values in the Value column**. Therefore, no imputation or estimation was required for the values used in the analysis.

3.5 Checking Data Quality

The dataset was checked for potential duplicate records and inconsistencies to ensure that observations could be correctly associated with their corresponding indicator and year.

3.6 Standardizing Data Types

The **Year** column was converted to a suitable numeric format, while the **Value** column was converted to a numeric data type to support calculations, descriptive analysis, and visualization.

3.7 Creating a Percentage Change Metric

To compare the long-term change across indicators, the percentage change between the earliest and latest available values for each indicator was calculated using:

$$\text{Percentage Change} = ((\text{Latest Value} - \text{Earliest Value}) / \text{Earliest Value}) \times 100$$

This metric was later used in Power BI to create a comparative visualization of long-term changes across the selected indicators.

4. Analysis Methodology

The project followed an end-to-end data analytics workflow.

4.1 Data Acquisition

The World Development Indicators dataset was obtained from the World Bank database.

4.2 Data Cleaning and Preparation

Python was used to filter the dataset to Nigeria, select relevant indicators, remove unnecessary information, assess missing values, and prepare the data for analysis.

4.3 Exploratory Data Analysis

Exploratory Data Analysis was conducted to examine:

- Dataset structure and dimensions.
- Data types.
- Descriptive statistics.
- Minimum and maximum values.
- Trends over time.
- Changes between the earliest and latest available observations.

4.4 Comparative Analysis

Percentage change was calculated for each indicator to allow comparison of long-term changes across indicators with different units of measurement.

4.5 Dashboard Development

Power BI was used to create an interactive dashboard containing:

- Five KPI cards.
- Population growth trend.
- GDP per capita trend.
- Electricity access trend.
- Forest area trend.
- Life expectancy trend.
- Percentage change comparison across indicators.
- Slicers for filtering the data.

4.6 Insight Generation

The results from the exploratory analysis and Power BI dashboard were interpreted to identify key development trends, areas of progress, and areas of concern.

5. Key Findings

The analysis identified several important trends in Nigeria's development indicators.

5.1 Significant Population Growth

Nigeria's population increased substantially from approximately **45 million in 1960** to over **237 million by 2025**.

This represents an increase of more than four times the country's population over the available period.

The increase in population has important implications for:

- Infrastructure demand.
- Housing.
- Healthcare.
- Education.
- Employment.
- Food security.
- Public services.

Population growth was one of the most significant long-term changes observed in the analysis.

5.2 Long-Term Increase in GDP per Capita

GDP per capita increased significantly over the period covered by the dataset, rising from approximately **\$93** in the earliest available observation to approximately **\$1,224** in 2025.

This represents substantial long-term nominal growth in GDP per capita.

However, GDP per capita should be interpreted alongside factors such as inflation, population growth, income distribution, and broader economic conditions. An increase in nominal GDP per capita does not necessarily mean that all citizens experienced the same improvement in living standards.

5.3 Improvement in Access to Electricity

Access to electricity increased from approximately **27% of the population in 1990** to approximately **62.5% in 2024**.

This represents significant progress in electricity access.

However, the data also shows that a substantial proportion of the population still lacked access to electricity in the latest available year.

This indicates that although progress has been made, electricity access remains an important development challenge.

5.4 Increase in Life Expectancy

Life expectancy increased from approximately **37 years in 1960** to approximately **55 years in 2024**.

This represents a long-term improvement in population health and survival outcomes.

The increase may reflect improvements in healthcare access, disease prevention, public health interventions, living conditions, and other social and economic factors.

Despite the improvement, the latest life expectancy figure indicates that continued investment in healthcare and living conditions remains important.

5.5 Decline in Forest Coverage

Forest area declined from approximately **29.1% of land area in 1990** to approximately **23.2% in 2023**.

This represents a decline of approximately **20.3%** relative to the earliest available value.

The decline indicates a reduction in forest coverage over the available period and raises concerns about:

- Deforestation.
- Habitat loss.
- Biodiversity.
- Soil degradation.

- Climate resilience.
- Environmental sustainability.

5.6 Increase in CO₂ Emissions per Capita

CO₂ emissions per capita increased slightly over the available period.

The percentage-change comparison showed an increase of approximately **9.8%**.

Although the increase was relatively small compared with some other indicators, it indicates additional environmental pressure associated with energy consumption, economic activity, and development.

5.7 Population Growth Remained Positive

Population growth remained positive over the available period.

Although the annual population growth rate was relatively small when expressed as a percentage, the long-term effect of sustained positive growth contributed to the substantial increase in Nigeria's total population.

6. Insights

The findings reveal several broader insights about Nigeria's development trajectory.

6.1 Demographic Growth Creates Increasing Development Pressure

The most significant overall trend was the substantial increase in Nigeria's population.

This means that improvements in infrastructure, healthcare, education, employment, and public services must keep pace with the increasing number of people requiring access to these services.

Economic growth and infrastructure expansion must therefore be evaluated in relation to population growth.

6.2 Development Progress Has Been Uneven

The analysis shows progress across several indicators, including:

- GDP per capita.
- Electricity access.
- Life expectancy.

However, environmental indicators moved in less positive directions, particularly forest coverage.

This suggests that economic and social development have not necessarily occurred without environmental costs.

6.3 Infrastructure Access Has Improved but Remains Incomplete

The increase in electricity access demonstrates significant progress.

However, with approximately 62.5% of the population having access to electricity in the latest available year, a considerable access gap remains.

This highlights the importance of continued investment in reliable and affordable electricity infrastructure.

6.4 Environmental Sustainability Requires Greater Attention

The decline in forest coverage is one of the clearest negative trends identified in the analysis.

The combination of declining forest area and increasing CO₂ emissions per capita suggests that environmental sustainability should remain an important part of Nigeria's development planning.

Economic development should therefore be pursued alongside policies that protect natural resources and reduce environmental degradation.

6.5 Improvements in Economic Indicators Should Be Interpreted Carefully

GDP per capita recorded one of the largest percentage increases across the indicators.

However, percentage change alone does not provide a complete measure of economic well-being.

The results should therefore be interpreted alongside population growth, access to basic infrastructure, health outcomes, inflation, and income distribution.

7. Recommendations

Based on the findings and insights from the analysis, the following recommendations are proposed.

7.1 Strengthen Infrastructure Development

Continued investment in electricity infrastructure is recommended to expand access to the remaining population without reliable electricity.

This should include:

- Expansion of electricity generation capacity.
- Improvement of transmission infrastructure.
- Improvement of distribution networks.
- Support for reliable off-grid and renewable energy solutions.

7.2 Plan Infrastructure Around Population Growth

Nigeria's substantial population growth requires long-term development planning that anticipates increasing demand for:

- Housing.
- Roads.
- Schools.
- Hospitals.
- Water supply.
- Electricity.
- Employment opportunities.

Population projections should be incorporated into infrastructure and public-service planning.

7.3 Promote Forest Conservation and Reforestation

The decline in forest coverage highlights the need for stronger environmental protection measures.

Recommended actions include:

- Reforestation programmes.
- Protection of existing forests.
- Sustainable land-use planning.
- Improved monitoring of deforestation.
- Enforcement of environmental regulations.
- Support for sustainable agricultural practices.

7.4 Promote Sustainable Economic Growth

Economic development should be pursued in a way that balances economic growth with environmental sustainability.

This could include:

- Investment in renewable energy.
- Improved energy efficiency.
- Cleaner production technologies.
- Sustainable transport systems.
- Policies aimed at reducing environmental degradation.

7.5 Continue Investment in Healthcare

The increase in life expectancy represents progress, but continued investment in healthcare remains important.

Priority areas should include:

- Access to primary healthcare.
- Preventive healthcare.

- Maternal and child health.
- Disease prevention.
- Healthcare infrastructure.
- Rural healthcare access.

7.6 Use Data-Driven Monitoring for Development Planning

The analysis demonstrates the value of monitoring development indicators over time.

Government agencies, development organisations, and policymakers should continue using regularly updated data to:

- Track progress.
 - Identify emerging challenges.
 - Evaluate policy outcomes.
 - Allocate resources effectively.
 - Support evidence-based decision-making.
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8. Conclusion

This project analyzed Nigeria's long-term development trends using selected indicators from the World Bank's World Development Indicators dataset.

The analysis revealed significant population growth, long-term increases in GDP per capita, improvements in electricity access, and increases in life expectancy. At the same time, the decline in forest coverage and increase in CO₂ emissions per capita highlight ongoing environmental challenges.

Overall, the findings demonstrate that Nigeria has experienced measurable progress across several dimensions of development. However, the results also show that development has been uneven and that economic and demographic progress must be accompanied by continued investment in infrastructure, healthcare, environmental protection, and sustainable development.

The Power BI dashboard provides an interactive way to explore these trends, while the percentage-change comparison offers a consolidated view of the long-term changes across the selected indicators.

Ultimately, the analysis demonstrates the importance of using data to understand development patterns and support informed decision-making. Nigeria's future development will require balancing population growth, economic opportunity, infrastructure expansion, social well-being, and environmental sustainability.

Project Deliverables

The completed project consists of:

- **Final Report:** This document.

- **Power BI Dashboard:** Interactive dashboard containing KPI cards, trend visualizations, comparative analysis, and slicers.
- **Power BI File:** `.pbix` project file.
- **Dashboard Screenshots:** Screenshots of the completed dashboard.
- **GitHub Repository:** Project files, documentation, and relevant analysis files.
- **Demo Video:** A walkthrough explaining the project objective, dataset, cleaning process, analysis, dashboard, key findings, and recommendations.